SEMESTER - III

COURSE: MAJOR – 3 (MJ-3) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: ELECTRICITY AND MAGNETISM **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objectives:

The course aims to develop a comprehensive understanding of electrostatics, dielectric and magnetic properties of matter, electromagnetic induction, and AC electrical circuits, emphasizing fundamental principles, mathematical formulations, and practical applications. It also aims to build problem-solving skills and a strong theoretical foundation for advanced studies in electromagnetism and applied physics.

Learning Outcomes:

1. Apply Gauss's law and solve Laplace's and Poisson's equations to analyze electric fields and potentials.
2. Understand conductor behavior, capacitance, and apply the method of images.
3. Examine dielectric and magnetic properties, including polarization, magnetization, and hysteresis.
4. Explain electromagnetic induction, and calculate self/mutual inductance and energy stored in magnetic fields.
5. Solve AC circuit problems using impedance, and analyze devices like bridges and transformers.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Electric Field and Electric Potential (15 Lectures): Electric flux, Gauss' law in integral and differential form and its applications, Conservative nature of Electrostatic Field, Laplace's and Poisson equations, The Uniqueness Theorem, Electric field and Potential due to electric dipole and quadrupole, Conductors in an electrostatic Field, Surface charge and force on a conductor, Capacitance of a system of charged conductors, Method of Images and its application to: (1) Plane Infinite Sheet and (2) Sphere.

UNIT-II:

Dielectric and Magnetic Properties of Matter (15 Lectures): Electric Field (E) in matter. Polarization (P) and Polarizability, Electrical susceptibility and Dielectric constant, Displacement vector D, Relations between E, P and D, Clausius-Mossotti equation, Gauss' Law in dielectrics, Magnetization vector (M), Magnetic field Intensity (H), Magnetic Susceptibility and permeability, Relation between B, H and M, B-H curve and hysteresis, Properties of magnetic materials- Dia, Para and Ferromagnetism, Langevin's theory, Measurement of susceptibility by Quincke's Method.

UNIT-III:

Electromagnetic Induction and Electrical Circuits (15 Lectures): Faraday's Law, Lenz's Law, Self-Inductance and Mutual Inductance, Reciprocity Theorem, Energy stored in a Magnetic Field, Charge Conservation and Displacement current, Torque on a current Loop, Ballistic Galvanometer: Current and Charge Sensitivity, Electromagnetic damping, Logarithmic damping. AC Circuits: Kirchhoff's laws for AC circuits, Complex Reactance and Impedance, Series LCR Circuit: (1) Resonance, (2) Power Dissipation and (3) Quality Factor, and (4) Band Width, Parallel LCR Circuit, Anderson's bridge, De-Sauty bridge and Carey Foster bridge, Equivalent circuit and vector diagram, Transformer, Losses in transformer.

SUGGESTED READINGS:

1. Electricity, Magnetism & Electromagnetic Theory, S. Mahajan and Choudhury, 2012, Tata McGraw.
2. Introduction to Electrodynamics, D.J. Griffiths, Cambridge University Press.
3. Feynman Lectures Vol.2, R. P. Feynman, R. B. Leighton, M. Sands, 2008, Pearson Education.
4. Electricity and Magnetism by K. K. Tewary, S Chand.
5. Electricity and Magnetism by D. C. Tayal, Himalaya Publishing House
6. Concepts of Physics-Classical Electromagnetism by H. C. Verma, BhartiBhawan.
7. Fundamentals of Magnetism and Electricity by D. N. Vasudeva, S. Chand.
8. Electrical circuits by M. R. Speigel, Schaum Series.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. To study the characteristics of a series RC Circuit.
2. To determine an unknown Low Resistance using Potentiometer.
3. To determine an unknown Low Resistance using Carey Foster's Bridge.
4. To compare capacitances using De'Sauty's bridge.
5. To determine self-inductance of a coil by Anderson's bridge.
6. To study response curve of a Series LCR circuit and determine its (a) Resonant frequency, (b) Impedance at resonance, (c) Quality factor Q, and (d) Band width.

SUGGESTED READINGS:

1. Introduction to Numerical Analysis, S.S. Sastry, 5th Edn. , 2012, PHI Learning Pvt. Ltd.
2. Advanced Practical Physics for students, B. L. Flint and H.T. Worsnop, 1971, Asia Publishing House.
3. A Text Book of Practical Physics, I.Prakash& Ramakrishna, 11th Edn, 2011, KitabMahal.
4. Numerical Methods, E Balagurusamy, McGraw Hill Education.

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SEMESTER - III

COURSE: MAJOR – 4(MJ-4) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01

PAPER NAME: Waves and Optics **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

Course Objective:

This course aims to provide students with a comprehensive understanding of wave phenomena in physics, including wave basics, wave optics, interference, diffraction, and polarization.

Learning Outcomes:

1. Understand the properties and behaviors of plane and spherical waves, longitudinal and transverse waves, and their mathematical representations.
2. Apply the wave equation and principles of energy transport to analyze wave propagation and intensity.
3. Analyze and interpret phenomena such as interference fringes, diffraction patterns, and polarization effects using theoretical models and experimental techniques.
4. Demonstrate proficiency in solving problems related to standing waves, interference, and diffraction in various mediums.
5. Explain the electromagnetic nature of light, including the laws of reflection and refraction, and apply them to optical systems such as lenses, mirrors, and interferometers.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

Wave Basics (15 Lectures): Plane and Spherical Waves, Longitudinal and Transverse Waves, Plane Progressive (Travelling) Waves, Wave Equation, Particle and Wave Velocities, Linearity and Superposition Principle, Superposition of two collinear oscillations having (1) equal frequencies and (2) different frequencies (Beats), Graphical and Analytical Methods, Lissajous Figures and their uses, : Standing

(Stationary) Waves in a String: Fixed and Free Ends, Analytical Treatment, Phase and Group Velocities, Changes with respect to Position and Time, Energy of Vibrating String, Transfer of Energy, Electromagnetic nature of light, Definition and properties of wave front, Huygens Principle, Temporal and Spatial Coherence, Fermat's Principle, Lens and Mirror formula, Laws of reflection and refraction, Cardinal points.

UNIT-II:

Interference and Diffraction (20 Lectures): Interference: Division of amplitude and wavefront, Interference in Thin Films, Fringes of equal inclination (Haidinger Fringes), Fringes of equal thickness (Fizeau Fringes), Newton's Rings: Measurement of wavelength, Measurement of refractive index. Michelson Interferometer, Michelson-Morley experiment and its failure, Determination of Wavelength, Wavelength Difference, Refractive Index, Visibility of Fringes, Fabry-Perot Interferometer.

Diffraction: Fresnel's Assumptions, Fresnel's half-Period Zones for Plane Wave, Explanation of Rectilinear Propagation of Light, Theory of a Zone Plate: Multiple Foci of a Zone Plate. Fresnel's Integral and its applications, Fresnel diffraction pattern of a straight edge, a slit and a wire. Fraunhofer Diffraction**:** Single slit diffraction, double slit diffraction, Circular aperture, Multiple slits Resolving Power of a telescope, Resolving power of grating, Use of grating to produce monochromatic light.

UNIT-III:

Polarization (10 Lectures): Polarization by reflection, Brewster's law, Double refraction, Nicol prism, Ordinary& extraordinary refractive indices, Retardation plate: $\lambda/2$ and $\lambda/4$ plates, Babinet compensator, Description of Linear, Circular and Elliptical Polarization, Production and detection of plane, circular, and elliptically polarized light. Optical activity.

SUGGESTED READINGS:

1. Waves: Berkeley Physics Course, vol. 3, Francis Crawford, 2007, Tata McGraw-Hill.
2. Fundamentals of Optics, F.A. Jenkins and H.E. White, 1981, McGraw-Hill
3. Principles of Optics, Max Born and Emil Wolf, 7th Edn., 1999, Pergamon Press.
4. Optics, Ajoy Ghatak, 2008, Tata McGraw Hill
5. The Physics of Vibrations and Waves, H. J. Pain, 2013, John Wiley and Sons.
6. The Physics of Waves and Oscillations, N.K. Bajaj, 1998, Tata McGraw Hill.
7. Optics by B. K. Mathur.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. Determine refractive index of the material of a prism using sodium source.
2. Determine the dispersive power and Cauchy constants of the material of a prism using mercury source.
3. Determine the wavelength of sodium source using Michelson's interferometer.
4. Determine the wavelength of sodium light using Newton's Rings.
5. Determine the radius of curvature of a plano-convex lens by using Newton's rings.
6. Determine the wavelength of (1) Na-source and (2) spectral lines of Hg-source using plane diffraction grating.

SUGGESTED READINGS:

1. Advanced Practical Physics for students, B.L. Flint and H.T. Worsnop, 1971, Asia Publishing House
2. A Text Book of Practical Physics, I. Prakash& Ramakrishna, 11th Ed., 2011, Kitab Mahal
3. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers
4. A Laboratory Manual of Physics for undergraduate classes, D.P. Khandelwal, 1985, Vani Pub.

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